

weighted least squares. Assuming that in (12) the variance is also proportional to the mean, the appropriate weights are given by $\exp(-x_i \tilde{\beta})$ and therefore the test can be performed by estimating

$$(y_i - \check{y}_i)^2 / \sqrt{\check{y}_i} = \lambda_0 \sqrt{\check{y}_i} + \lambda_0 (\lambda_1 - 1) \ln(\check{y}_i) \sqrt{\check{y}_i} + \xi_i^* \quad (13)$$

by OLS and testing the statistical significance of $\lambda_0 (\lambda_1 - 1)$ using a Eicker-White robust covariance matrix estimator.¹⁵

In the next section, a small simulation is used to study the Gauss-Newton regression test for the hypothesis that $V[y_i|x] \propto E[y_i|x]$, as well as the Park-type test for the hypothesis that the constant-elasticity model can be consistently estimated in the log-linear form.

4. A simulation study

This section reports the results of a small simulation study designed to assess the performance of different methods to estimate constant-elasticity models in the presence of heteroskedasticity and rounding errors. As a by-product, we also obtain some evidence on the finite sample performance of the specification tests presented above. These experiments are centered around the following multiplicative model:

$$E[y_i|x] = \mu(x_i \beta) = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}), \quad i = 1, \dots, 1000. \quad (14)$$

Since, in practice, regression models often include a mixture of continuous and dummy variables, we replicate this feature in our experiments: x_{1i} is drawn from a standard normal and x_2 is a binary dummy variable that equals 1 with a probability of 0.4.¹⁶ The

¹⁵Notice that to test $V[y_i|x] \propto E[y_i|x]$ against alternatives of the form $V[y_i|x] = \lambda_0 \exp(x_i(\beta + \lambda))$ the appropriate auxiliary regression would be

$$(y_i - \check{y}_i)^2 / \sqrt{\check{y}_i} = \lambda_0 \sqrt{\check{y}_i} + \lambda_0 \lambda x_i \sqrt{\check{y}_i} + \xi_i^*,$$

and the test could be performed by checking the joint significance of the elements of $\lambda_0 \lambda$. If the model includes a constant, one of the regressors in the auxiliary regression is redundant and should be dropped.

¹⁶For example, in gravity equations, continuous variables (which are all strictly positive) include income and geographical distance. In equation (14), x_1 can be interpreted as (the logarithm of) one of these

two covariates are independent and a new set of observations of all variables is generated in each replication using $\beta_0 = 0$, $\beta_1 = \beta_2 = 1$. Data on y are generated as

$$y_i = \mu(x_i\beta) \eta_i, \quad (15)$$

where η_i is a log-normal random variable with mean 1 and variance σ_i^2 . As noted before, the slope parameters in (14) can be estimated using the log-linear form of the model only when σ_i^2 is constant. That is, when $V[y_i|x]$ is proportional to $\mu(x_i\beta)^2$.

In these experiments we analyzed PML estimators of the multiplicative model and different estimators of the log-linearized model. The consistent PML estimators studied were: NLS, gamma pseudo-maximum likelihood (GPML) and the PPML. Besides these estimators, we also considered the standard OLS estimator of the log-linear model, OLS; the OLS estimator for the model where the dependent variable is $y_i + 1$, OLS ($y + 1$); a truncated OLS estimators to be discussed below; and the threshold Tobit of Eaton and Tamura (1994), ET-Tobit.¹⁷

To assess the performance of the estimators under different patterns of heteroskedasticity, we considered the four following specifications of σ_i^2 :

Case 1: $\sigma_i^2 = \mu(x_i\beta)^{-2}$; $V[y_i|x] = 1$;

Case 2: $\sigma_i^2 = \mu(x_i\beta)^{-1}$; $V[y_i|x] = \mu(x_i\beta)$;

Case 3: $\sigma_i^2 = 1$; $V[y_i|x] = \mu(x_i\beta)^2$;

Case 4: $\sigma_i^2 = \exp(x_{2i}) + \mu(x_i\beta)^{-1}$; $V[y_i|x] = \mu(x_i\beta) + \exp(x_{2i}) \mu(x_i\beta)^2$.

In Case 1 the variance of ε_i is constant, implying that the NLS estimator is optimal. Although, as argued before, this case is unrealistic for models of bilateral trade, it is included in the simulations for completeness. In Case 2, the conditional variance of y_i equals its conditional mean, as in the Poisson distribution. The pseudo-likelihood variables. Examples of binary variables include dummies for free-trade agreements, common language, colonial ties, contiguity and access to land.

¹⁷We also studied the performance of other variants of the Tobit model, finding very poor results.

estimator based on the Poisson distribution is optimal in this situation. Case 3 is the special case in which OLS estimation of the log-linear model is consistent for the slope parameters of (14). Moreover, in this case the log-linear model not only corrects the heteroskedasticity in the data, but, because η_i is log-normal, it is also the maximum likelihood estimator. The GPML is the optimal PML estimator in this case, but it should be outperformed by the true maximum likelihood estimator. Finally, Case 4 is the only one in which the conditional variance does not depend exclusively on the mean. The variance is a quadratic function of the mean, as in Case 3, but it is not proportional to the square of the mean.

We carried out two sets of experiments. The first set was aimed at studying the performance of the estimators of the multiplicative and the log-linear models under different patterns of heteroskedasticity. In order to study the effect of the truncation on the performance of the OLS, and given that this data generating mechanism does not produce observations with $y_i = 0$, the log-linear model was also estimated using only the observations for which $y_i > 0.5$, OLS ($y > 0.5$). This reduces the sample size by about 25% to 35%, depending on the pattern of heteroskedasticity. The estimation of the threshold Tobit was also performed using this dependent variable. Notice that, although the dependent variable has to cross a threshold to be observable, the truncation mechanism used here is not equal to the one assumed by Eaton and Tamura (1994). Therefore, in all these experiments the ET-Tobit will be slightly misspecified and the results presented here should be viewed as a check of its robustness to this problem.

The second set of experiments studied the estimators' performance in the presence of rounding errors in the dependent variable. For that purpose, a new random variable was generated rounding to the nearest integer the values of y_i obtained in the first set of simulations. This procedure mimics the rounding errors in official statistics and generates a large number of zeros, a typical feature of trade data. Because the model considered here generates a large proportion of observations close to zero, rounding down is much more frequent than rounding up. As the probability of rounding up or down depends on the covariates, this procedure will necessarily bias the estimates, as discussed before. The

purpose of the study is to gauge the magnitude of these biases. Naturally, the log-linear model cannot be estimated in these conditions because the dependent variable equals zero for some observations. Following what is the usual practice in these circumstances, the truncated OLS estimation of the log-linear model was performed dropping the observations for which the dependent variable equals zero. Notice that the observations discarded with this procedure are exactly the same that are discarded by OLS ($y > 0.5$) in the first set of experiments. Therefore, this estimator is also denoted OLS ($y > 0.5$).

The results of the two sets of experiments are summarized in Table 1, which displays the biases and standard errors of the different estimators of β obtained with 10,000 replicas of the simulation procedure described above. Only results for β_1 and β_2 are presented as these are generally the parameters of interest.

As expected, OLS only performs well in Case 3. In all other cases this estimator is clearly inadequate because, despite its low dispersion, it is often badly biased. Moreover, the sign and magnitude of the bias vary considerably. Therefore, even when the dependent variable is strictly positive, estimation of constant elasticity models using the log-linearized model cannot generally be recommended. As for the modifications of the log-linearized model designed to deal with the zeros of the dependent variable, ET-Tobit, OLS ($y + 1$) and OLS ($y > 0.5$), their performance is also very disappointing. These results clearly emphasize the need to use adequate methods to deal with the zeros in the data and raise serious doubts about the validity of the results obtained using the traditional estimators based on the log-linear model. Overall, except under very special circumstances, estimation based on the log-linear model cannot be recommended.

One remarkable result of this set of experiments is the extremely poor performance of the NLS estimator. Indeed, when the heteroskedasticity is more severe (cases 3 and 4) this estimator, despite being consistent, leads to very poor results because of its erratic behavior.¹⁸ Therefore, it is clear that the loss of efficiency caused by some of the forms of heteroskedasticity considered in these experiments is strong enough to render this estimator useless in practice.

¹⁸Manning and Mullahy (2001) report similar results.